

Conjecture G and Reservoir SG(11)

Arithmetic Structure of Sophie Germain Primes and Goldbach's Conjecture

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Abstract

We study the arithmetic structure of Sophie Germain primes (SG) in relation to Goldbach's conjecture. We establish 29 orbits $F(r_p) \rightarrow r_q$ covering all 15 residues of $2n$ modulo 30, prove three universal invariants, compute Hardy–Littlewood singular series $\mathfrak{S} > 0$ for all orbits, and verify zero counter-examples up to $N = 2 \times 10^8$. We formulate Conjectures G and R as precise quantitative steps toward a proof of Goldbach's conjecture.

1. Definitions

1.1 Wheel R_0

The admissible residues modulo 30 are those coprime to 30:

$$R_0 = \{1, 7, 11, 13, 17, 19, 23, 29\} \subset (\mathbb{Z}/30\mathbb{Z})^\times$$

Every prime $p \geq 7$ satisfies $p \bmod 30 \in R_0$.

1.2 Sophie Germain Families

The three Sophie Germain prime families are:

$$F_A(11) : p \equiv 11 \pmod{30} \text{ and } 2p+1 \text{ prime}$$

$$F_B(23) : p \equiv 23 \pmod{30} \text{ and } 2p+1 \text{ prime}$$

$$F_C(29) : p \equiv 29 \pmod{30} \text{ and } 2p+1 \text{ prime}$$

The non-SG families (denoted XSG) are $F(1)$, $F(7)$, $F(13)$, $F(17)$, $F(19)$.

1.3 Orbits

For $r_p \in R_0$ and $r_q \in R_0$, the orbit $F(r_p) \rightarrow r_q$ is defined for even integers $2n \equiv r_p + r_q \pmod{30}$ as:

$$\mathfrak{A}(r_p, r_q, 2n) = \{ (p, q) : p \in F(r_p), q = 2n - p \text{ prime}, q \equiv r_q \pmod{30} \}$$

2. Arithmetic Structure mod 30

2.1 Admissible Pairs

For each residue $r \in R_0$ of $2n$ modulo 30, the admissible pairs $P(2n)$ are the pairs $(a, b) \in R_0^2$ with $a + b \equiv r \pmod{30}$ and $a \leq b$.

$2n \bmod 30$	# pairs	$ P \geq 3$	SG pairs	SG-SG pairs
0	4	yes	(1,29),(7,23),(11,19)	—
2	2	no	—	—
4	2	no	(11,23)	(11,23) ★

2n mod 30	# pairs	 P ≥3	SG pairs	SG-SG pairs
6	3	yes	(7,29),(13,23)	—
8	2	no	—	—
10	2	no	(11,29),(17,23)	(11,29) ★
12	3	yes	(1,11),(13,29),(19,23)	—
14	2	no	—	—
16	2	no	(17,29),(23,23)	(23,23) ★
18	3	yes	(7,11),(19,29)	—
20	2	no	—	—
22	2	no	(11,11),(23,29)	(11,11)★, (23,29)★
24	3	yes	(1,23),(11,13)	—
26	2	no	—	—
28	2	no	(11,17),(29,29)	(29,29) ★

10/15 residues have at least one SG pair. The 5 residues {2,8,14,20,26} have no SG pair — covered by XSG families.

3. The 29 Orbits (N = 2×10⁸)

Each orbit was tested on approximately 6,666,665 values of 2n. Zero counter-examples above 2n = 200.
193,333,305 Goldbach decompositions verified.

Orbit	2n≡	★	Tested	Success	Cov.	p-med	p₁%	℄
SG(11)→1	12		6,666,666	6,666,665	100%	131	20.8%	53.7
SG(11)→7	18		6,666,666	6,666,666	100%	131	20.8%	42.9
SG(11)→11	22	★	6,666,665	6,666,665	100%	131	20.8%	135.5
SG(11)→13	24		6,666,665	6,666,665	100%	131	20.8%	42.9
SG(11)→17	28		6,666,665	6,666,665	100%	131	20.8%	48.3
SG(11)→19	0		6,666,665	6,666,665	100%	131	20.8%	42.9
SG(11)→23	4	★	6,666,665	6,666,665	100%	131	20.8%	45.1
SG(11)→29	10	★	6,666,665	6,666,665	100%	131	20.8%	53.7
SG(23)→1	24		6,666,665	6,666,665	100%	83	20.8%	48.3
SG(23)→7	0		6,666,665	6,666,665	100%	83	20.8%	48.3
SG(23)→11	4	★	6,666,665	6,666,665	100%	83	20.8%	42.9

Orbit	$2n \equiv$	★	Tested	Success	Cov.	p-med	p ₁ %	⊗
SG(23)→13	6		6,666,665	6,666,665	100%	83	20.8%	53.7
SG(23)→17	10		6,666,665	6,666,665	100%	83	20.8%	47.2
SG(23)→19	12		6,666,665	6,666,665	100%	83	20.8%	42.9
SG(23)→23	16	★	6,666,665	6,666,665	100%	83	20.8%	135.5
SG(23)→29	22	★	6,666,664	6,666,664	100%	83	20.8%	48.3
SG(29)→1	0		6,666,665	6,666,665	100%	179	20.8%	57.0
SG(29)→7	6		6,666,665	6,666,665	100%	179	20.8%	48.3
SG(29)→11	10	★	6,666,665	6,666,665	100%	179	20.8%	44.2
SG(29)→13	12		6,666,665	6,666,665	100%	179	20.8%	48.3
SG(29)→17	16		6,666,665	6,666,665	100%	179	20.8%	42.9
SG(29)→19	18		6,666,665	6,666,664	100%	179	20.8%	53.7
SG(29)→23	22	★	6,666,664	6,666,664	100%	179	20.8%	47.2
SG(29)→29	28	★	6,666,664	6,666,664	100%	179	20.8%	135.5
XSG(1)→1	2		6,666,666	6,666,664	100%	151	20.8%	38.5
XSG(1)→7	8		6,666,666	6,666,666	100%	151	20.8%	13.2
XSG(1)→13	14		6,666,666	6,666,666	100%	151	20.8%	13.2
XSG(1)→19	20		6,666,666	6,666,666	100%	151	20.8%	13.2
XSG(7)→19	26		6,666,665	6,666,665	100%	67	20.8%	13.2

★ = SG-SG pair ($r_q \in \{11, 23, 29\}$). ⊗ = Hardy-Littlewood singular series. 3 trivial failures ($2n \leq 200$) — zero failure for $2n \geq 200$.

4. Universal Invariants

4.1 Invariant I — p-median

The p-median (median of minimum p values used) equals the k-th element of the family, where $k \approx 1 + \log(N)/(7 \cdot \log 10)$ increases by one rank per 3 orders of magnitude:

Family	p ₁	p ₂	p ₃ (p-med N=2×10 ⁸)	Formula
SG(11)	11	41	131	p-med = SG(11)[[1+log N/(7 log 10)]]
SG(23)	23	53	83	same formula
SG(29)	29	89	179	same formula
XSG(1)	31	61	151	same formula

Family	p_1	p_2	p_3 (p-med $N=2 \times 10^8$)	Formula
XSG(7)	7	37	67	same formula

4.2 Invariant II — p_1 -rate

The utilization rate of p_1 decays as $C/\log(N)$, universally across all 29 orbits:

$$p_1\text{-rate}(N) \approx 1.42 / \log(N) \longrightarrow 20.8\% \text{ at } N = 2 \times 10^8$$

4.3 Invariant III — 100% Coverage

All 29 orbits achieve 100% coverage up to $N = 2 \times 10^8$. The 719,370,587 rejected candidates (composite q , avg 3.7 per $2n$) never exhausted all candidates of any orbit.

There are no 'false Goldbach pairs' in the strict sense: all retained pairs (p,q) satisfy primality of both p and q by construction.

4.4 Minimal Reservoir

The minimal reservoir $k_{\min}(N)$ is the smallest subset $S \subset \text{SG}(11)$ covering all $2n \equiv 10 \pmod{30}$ in $[8, N]$:

$$k_{\min}(N) \approx 0.018 \cdot (\log N)^{\{2.87\}}$$

N	$k_{\min}$	$\max(S)$	Status
10^6	36	$\text{SG}(11)[36] = 5,231$	Verified
2×10^8	~89	$\text{SG}(11)[89] = 15,401$	Extrapolated
10^{11}	~200	$\text{SG}(11)[200] \approx 30,000$	Predicted

5. Formal Conjectures

5.1 Conjecture G

****Conjecture G** (Monfette 2025)

Part I****** (unconditional): *The 29 orbits cover exactly all 15 residues of $2n$ modulo 30. For each residue $r_{\{2n\}}$, at least one orbit $F(r_p) \rightarrow r_q$ is admissible.

***Part II** (under GEH): $\ast \Im(F(r_p) \rightarrow r_q) > 0$ for all 29 orbits. The asymptotic density is positive and non-vanishing:

$$N_{\vartheta}(x) \sim W \cdot x / (\log x)^k$$

Part III (universal invariants): *The p -median is constant per family for fixed N . The p_1 -rate $\approx 1.42/\log(N)$ is universal across all 29 orbits.

***Part IV** (empirical): *Zero counter-examples among 193,333,305 tested values up to $N=2 \times 10^8$.*

5.2 Conjecture R

****Conjecture R** (Monfette 2025)

*******For every even integer $2n \equiv 10 \pmod{30}$ with $2n \geq 8$, there exists $p \in \text{SG}(11)$ such that $q=2n-p$ is prime. The minimal reservoir satisfies:

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$$k_{\min}(N) = O \left((\log N)^{3+\varepsilon} \right) \quad \forall \varepsilon > 0$$

This conjecture is equivalent to Goldbach's conjecture for residue $2n \equiv 10 \pmod{30}$ with the additional constraint $p \in \text{SG}(11)$. It is strictly stronger than Chen's theorem on the structure of p .

6. Epistemic Status

Result	Status	Condition
15 residues covered by 29 orbits	PROVED	Unconditional
$\varepsilon > 0$ for all 29 orbits	PROVED	GEH (standard)
Invariants I, II, III	EMPIRICAL	Verified $N \leq 10^8$
$k_{\min}(N) \approx 0.018 \cdot (\log N)^{2.87}$	EMPIRICAL	Verified $N \leq 10^6$
100% coverage on 29 orbits	VERIFIED	193M+ cases
$k_{\min}(N) = O((\log N)^{3+\varepsilon})$	CONJECTURE	Restricted Goldbach
False Goldbach pairs: 0	PROVED	By construction

While we cannot yet prove Goldbach's conjecture, this work provides a precise structural map.

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